ESTIMACIÓN DEL PRECIO DE VENTA DE VEHÍCULOS USADOS EN LA INDIA

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DEPURACIÓN DEL DATASET

- **RECODIFICACIÓN DE LAS VARIABLES**

  make (fabricante del coche), *registered_city* (ciudad en la que está registrado el coche)

- **ERROR EN LA VARIABLE OBJETIVO**

  sale_price

- **DATOS ATÍPICOS**

  i. Porcentaje alto de atípicos ⟶ distribución rara

  ii. Porcentaje alto de atípicos ⟶ borrarlos

- **DATOS AUSENTES**

  Porcentaje muy pequeño ⟶ los quitamos

MODELO GLM

1º Modelo inicial y ANOVA

modeloInicial ← lm(sale_price~., data=datos_train)

```
Anova Table (Type II tests)

Response: sale_price
                      Sum Sq   Df   F value    Pr(>F)
yr_mfr              3.8797e+13    1 1875.4759 < 2.2e-16 ***
fuel_type           8.3097e+12    4  100.4239 < 2.2e-16 ***
kms_run             1.3095e+12    1   63.2996 2.149e-15 ***
body_type           4.0574e+13    4  490.3442 < 2.2e-16 ***
transmission        6.9810e+12    1  337.4678 < 2.2e-16 ***
assured_buy         7.5423e+10    1    3.6460 0.0562579 .
is_hot              2.8943e+11    1   13.9914 0.0001856 ***
total_owners        4.2098e+11    1   20.3503 6.586e-06 ***
car_rating          2.8634e+11    3    4.6139 0.0031530 **
fitness_certificate 7.0482e+10    1    3.4071 0.0649705 .
warranty_avail      1.2995e+10    1    0.6282 0.4280526
casa                2.3970e+13   13   89.1341 < 2.2e-16 ***
region              5.9306e+12    4   71.6725 < 2.2e-16 ***
year_anuncio        6.3128e+11    1   30.5164 3.466e-08 ***
mes_anuncio         3.5697e+11    1   17.2562 3.317e-05 ***
dia_anuncio         2.4781e+11    1   11.9794 0.0005421 ***
Residuals           1.1105e+14 5368
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

2º Modelo con las variables significativas “modelo2”

```{r}
add1(modelo2, scope=formula(modeloInicial),test = "F")
```

```
Single term additions

Model:
sale_price ~ (yr_mfr + fuel_type + kms_run + body_type + transmission +
    assured_buy + is_hot + total_owners + car_rating + fitness_certificate +
    warranty_avail + casa + region + year_anuncio + mes_anuncio +
    dia_anuncio) - warranty_avail - fitness_certificate - assured_buy
                    Df  Sum of Sq        RSS    AIC F value  Pr(>F)
<none>                            1.1122e+14 128497
assured_buy          1 9.0383e+10 1.1113e+14 128495  4.3675 0.03668 *
fitness_certificate  1 7.8772e+10 1.1114e+14 128495  3.8060 0.05112 .
warranty_avail       1 2.3429e+10 1.1120e+14 128498  1.1314 0.28752
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

modelo2Def

```{r}
modelo2Def <- lm(sale_price~.-warranty_avail-fitness_certificate, data=datos_train)
R2(predict(modelo2Def,datos_train), datos_train$sale_price)
```

```
[1] 0.7062125
```

```{r}
R2(predict(modelo2Def,datos_test), datos_test$sale_price)
```

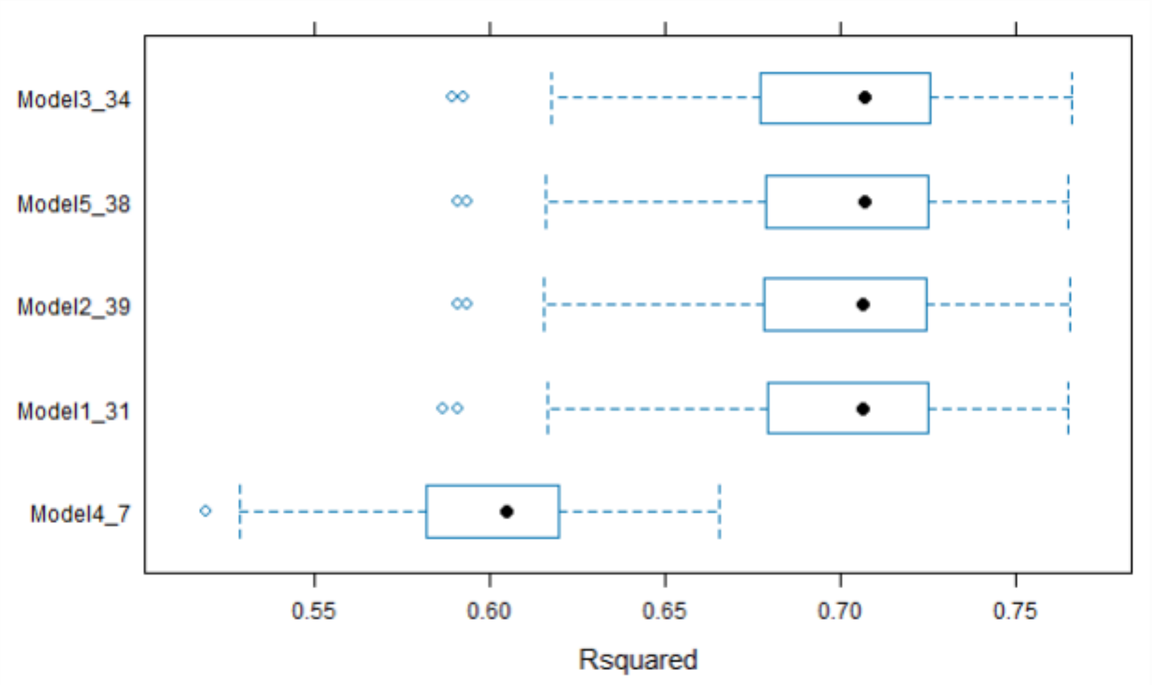
```
[1] 0.7032461
```

```{r}
modelo2Def$rank
```

```
[1] 38
```

MODELO GLM

3º Métodos automáticos de selección de variables → **modeloStepBIC**




```{r}
summary(resamples(vcrTodosModelos), metric=c("Rsquared"))
```

```
Call:
summary.resamples(object = resamples(vcrTodosModelos), metric = c("Rsquared"))

Models: Model1_31, Model2_39, Model3_34, Model4_7, Model5_38
Number of resamples: 100

Rsquared
               Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
Model1_31 0.5870418 0.6793404 0.7063565 0.6975105 0.7250148 0.7648312    0
Model2_39 0.5911640 0.6787563 0.7067381 0.6981383 0.7244477 0.7654237    0
Model3_34 0.5894424 0.6778623 0.7073730 0.6983136 0.7251605 0.7660665    0
Model4_7  0.5191709 0.5826463 0.6049905 0.6008428 0.6194909 0.6653970    0
Model5_38 0.5911704 0.6793927 0.7072186 0.6986647 0.7249539 0.7653192    0
```

MODELO GLM

modeloStepBIC:

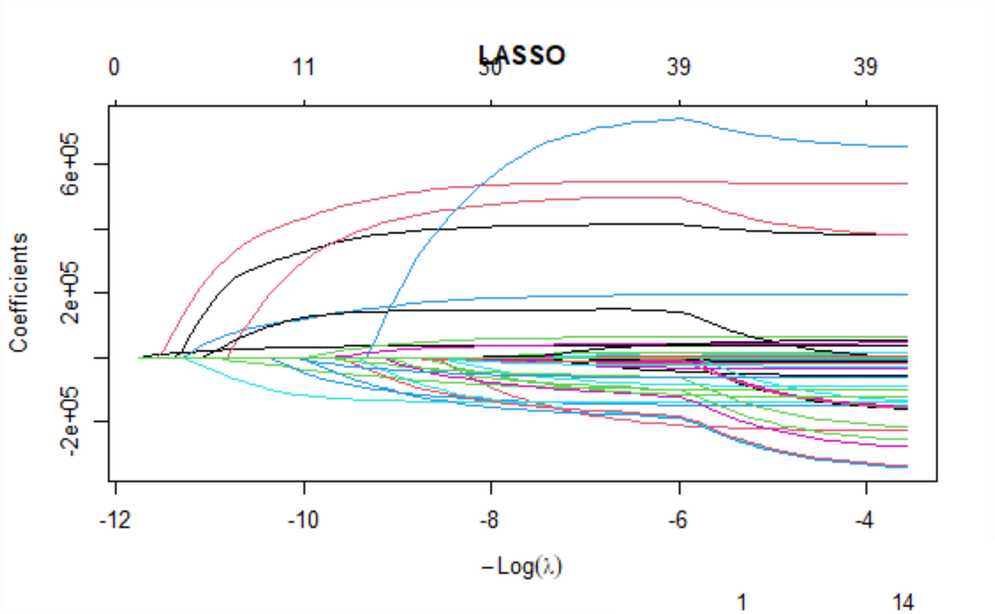
- **TRAIN**: 0.7034
- **TEST**: 0.7043
- **RANK**: 31

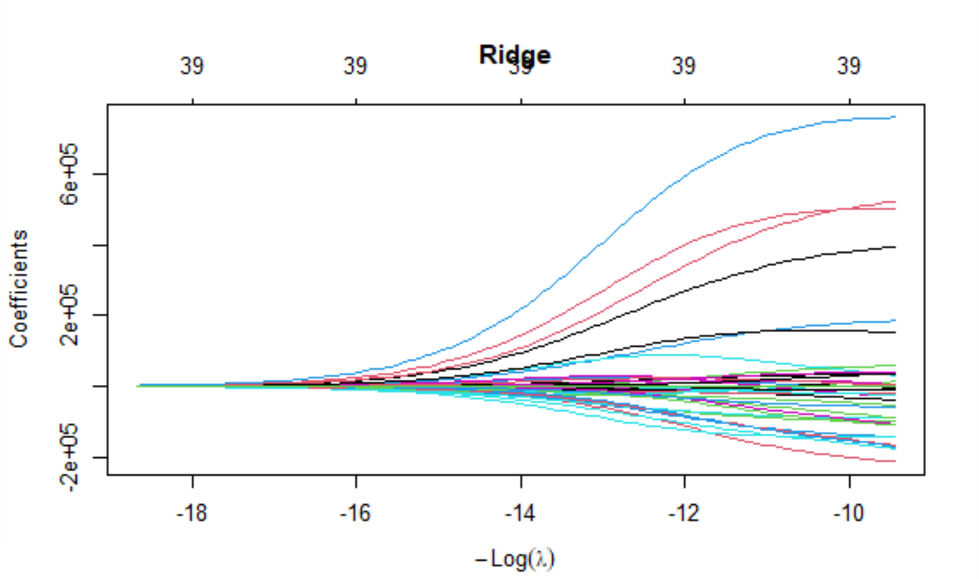
```{r}
Anova(modeloStepBIC,type = "II")
```

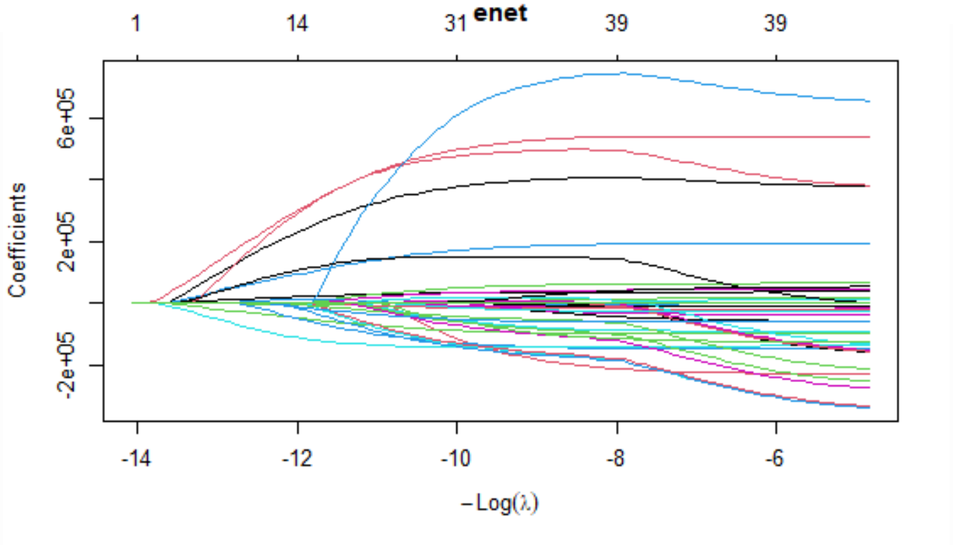
```
Anova Table (Type II tests)

Response: sale_price
                Sum Sq   Df  F value    Pr(>F)
body_type    4.0875e+13    4  489.732 < 2.2e-16 ***
yr_mfr       4.1968e+13    1 2011.303 < 2.2e-16 ***
casa         2.3566e+13   13   86.878 < 2.2e-16 ***
transmission 7.0040e+12    1  335.668 < 2.2e-16 ***
fuel_type    7.9979e+12    4   95.825 < 2.2e-16 ***
region       6.3196e+12    4   75.717 < 2.2e-16 ***
kms_run      1.3838e+12    1   66.318 4.730e-16 ***
total_owners 3.7908e+11    1   18.167 2.058e-05 ***
dia_anuncio  1.8274e+11    1    8.758  0.003096 **
Residuals    1.1220e+14 5377
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)           -7.469e+07  1.686e+06 -44.289  < 2e-16 ***
body_typeluxury sedan  3.759e+05  1.863e+04  20.181  < 2e-16 ***
body_typeluxury suv    5.374e+05  1.422e+04  37.803  < 2e-16 ***
body_typesedan         6.591e+04  5.739e+03  11.484  < 2e-16 ***
body_typesuv           1.960e+05  6.931e+03  28.277  < 2e-16 ***
yr_mfr                 3.750e+04  8.363e+02  44.848  < 2e-16 ***
casaDaimler            3.579e+05  3.979e+04   8.994  < 2e-16 ***
casaFord              -2.742e+05  3.331e+04  -8.233 2.28e-16 ***
casaGeely_Group        6.380e+05  1.474e+05   4.330 1.52e-05 ***
casaGM                -3.463e+05  3.858e+04  -8.974  < 2e-16 ***
casaHonda_Group       -1.760e+05  3.191e+04  -5.514 3.66e-08 ***
casaHyundai_Group     -1.827e+05  3.124e+04  -5.846 5.32e-09 ***
casaNissan_Alliance   -3.528e+05  3.582e+04  -9.848  < 2e-16 ***
casaOtros             -2.372e+05  3.110e+04  -7.626 2.84e-14 ***
casaRenault_Group     -3.605e+05  3.296e+04 -10.937  < 2e-16 ***
casaStellantis        -1.553e+05  4.382e+04  -3.544 0.000398 ***
casaTata_Motors       -2.889e+05  3.402e+04  -8.492  < 2e-16 ***
casaToyota_Group      -2.201e+04  3.123e+04  -0.705 0.481048
casaVolkswagen_Group  -1.782e+05  3.134e+04  -5.686 1.37e-08 ***
transmissionmanual    -1.347e+05  7.353e+03 -18.321  < 2e-16 ***
fuel_typeelectric     -2.184e+05  7.263e+04  -3.007 0.002654 **
fuel_typepetrol       -9.420e+04  5.592e+03 -16.844  < 2e-16 ***
fuel_typepetrol & cng -1.430e+05  9.519e+03 -15.022  < 2e-16 ***
fuel_typepetrol & lpg -7.961e+04  2.798e+04  -2.845 0.004458 **
regionEste            -1.015e+05  3.034e+04  -3.347 0.000824 ***
regionNorte           -6.259e+04  4.824e+03 -12.975  < 2e-16 ***
regionOeste           -2.581e+04  7.111e+03  -3.630 0.000286 ***
regionSur              4.404e+04  6.023e+03   7.312 3.03e-13 ***
kms_run               -4.839e-01  5.942e-02  -8.144 4.73e-16 ***
total_owners          -1.620e+04  3.801e+03  -4.262 2.06e-05 ***
dia_anuncio            6.732e+02  2.275e+02   2.959 0.003096 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

MODELO GLM - REGRESIÓN PENALIZADA

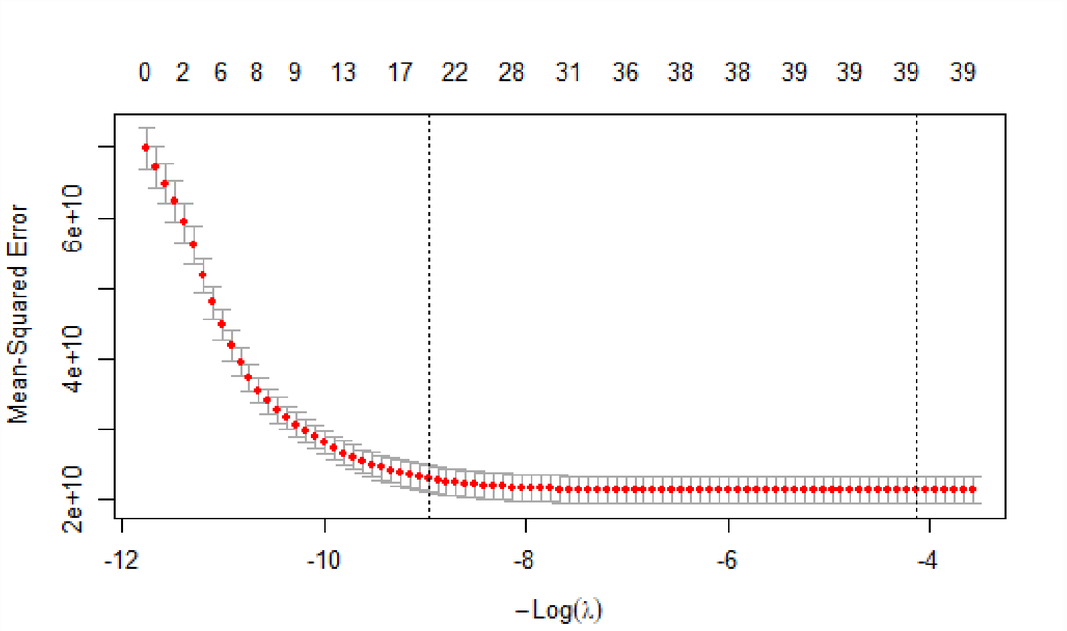




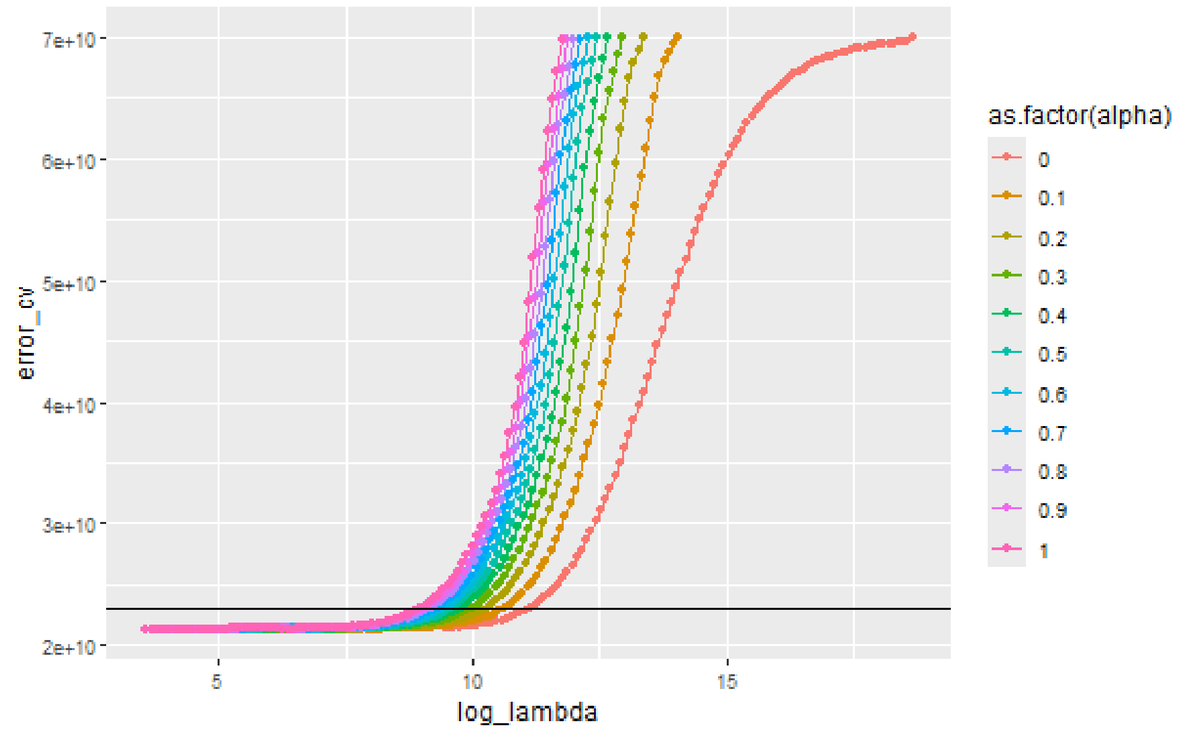


MODELO GLM - REGRESIÓN PENALIZADA

1º Evolución del ECM en base a la penalización:



2º Evolución del ECM en base a los alphas:



MODELO GLM - REGRESIÓN PENALIZADA

3 combinaciones de parámetros que dan lugar a modelos con 20 parámetros no nulos

	alpha <dbl>	lambda <dbl>	log_lambda <dbl>	numPar <int>	error_cv <dbl>	error_sd <dbl>	error_up <dbl>
s30...1	1.0	7843.857	8.967486	20	23042074404	2191400185	25233474589
s30...2	0.9	8715.396	9.072846	20	23064730528	2194228947	25258959475
s30...3	0.8	9804.821	9.190629	20	23093811185	2197824791	25291635976

La combinación seleccionada de α=1 y λ=7843.857 obtiene un R2:

- **TRAIN**: 0.6848
- **TEST**: 0.6796

Frente a los resultados del GLM anterior:

- **TRAIN**: 0.7034
- **TEST**: 0.7043
- **RANK**: 31

modeloStepBIC

MODELO DE ÁRBOL

1º Creación de los modelos de árbol

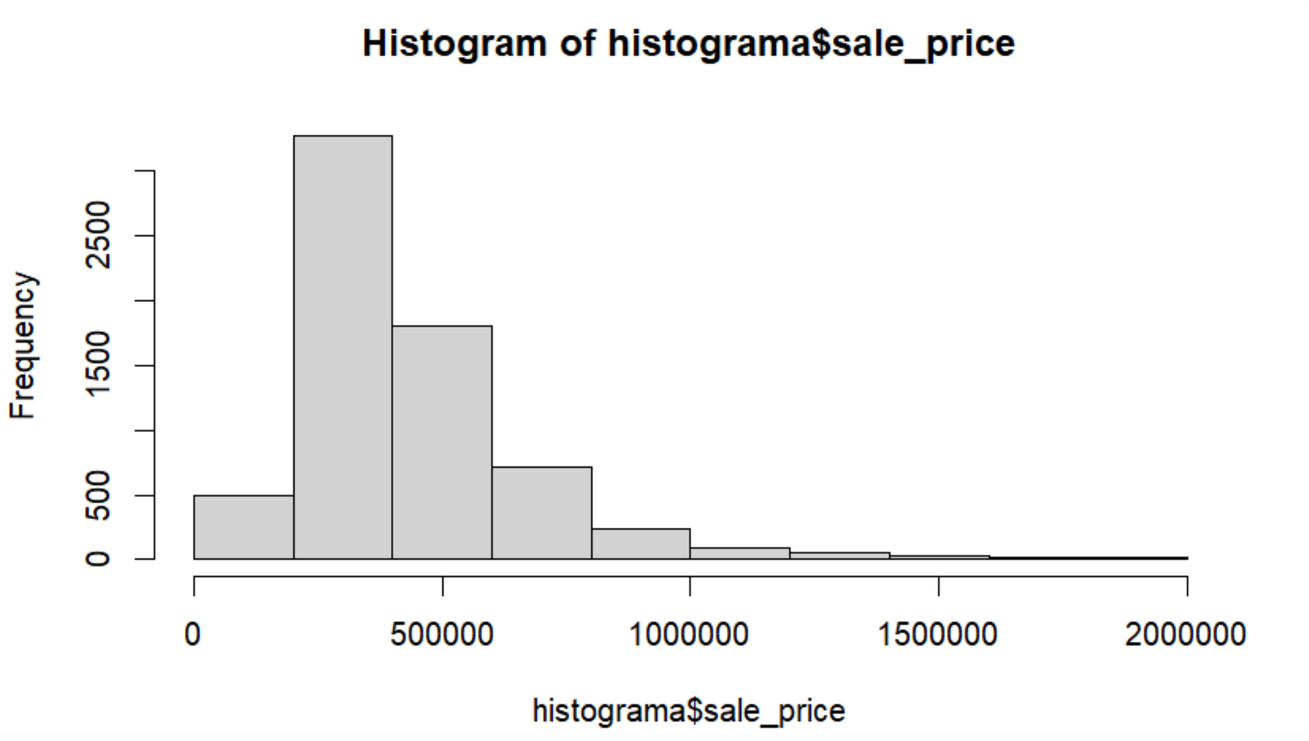
2º Selección del modelo óptimo

3º Poda del árbol

4º Árbol Podado

5º Importancia de las variables

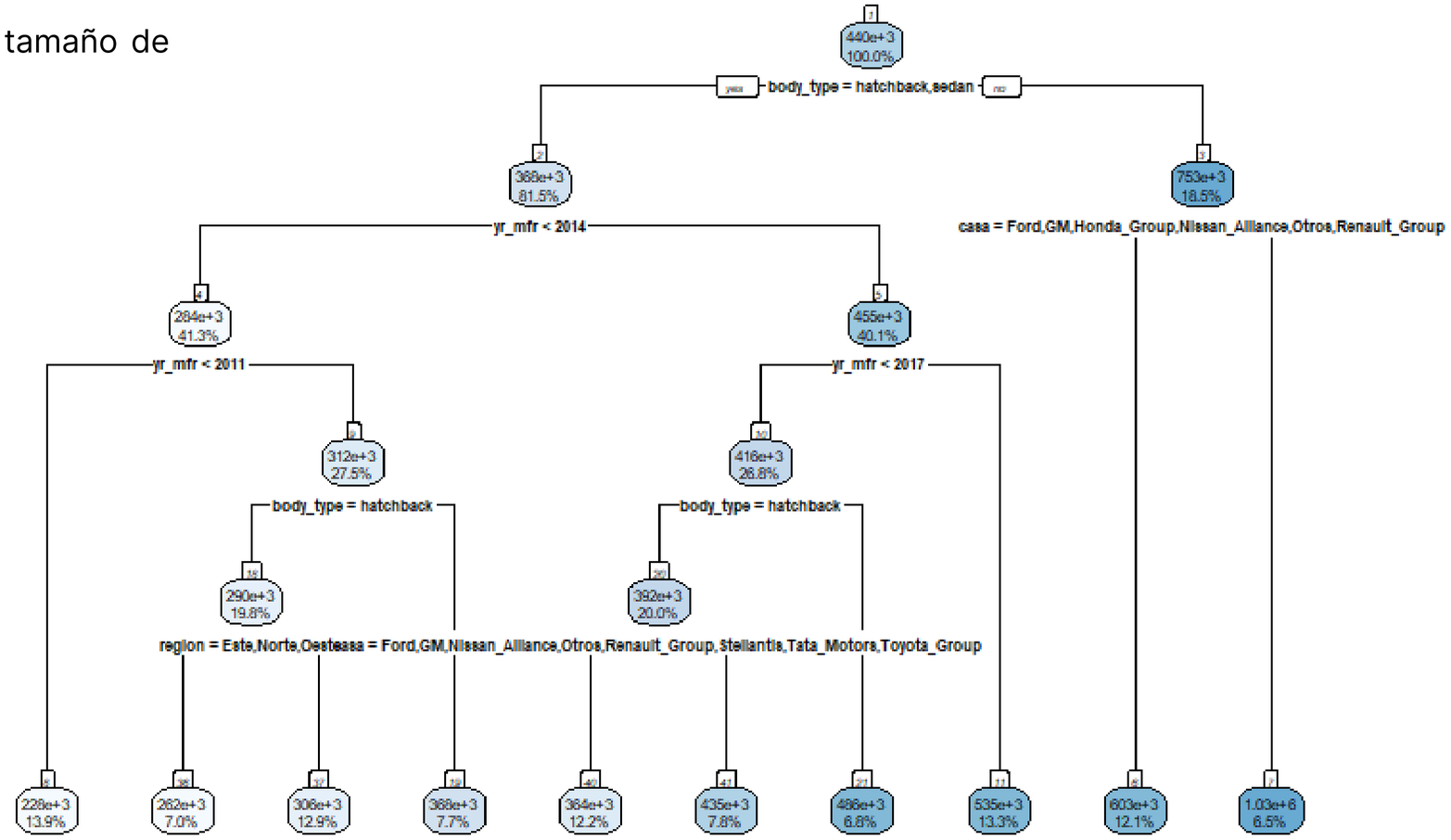
6º Gestión de ausentes



CREACIÓN DE LOS ÁRBOLES

- Los modelos creados han sido con un tamaño de hoja del 5%, 2%, 1% y 0,5%.
- Número de hojas: 10, 27, 53 y 106
- Profundidad: 5, 7, 10, 13

Ejemplo del modelo con 5%



SELECCIÓN DEL ÁRBOL

	Modelo 5	Modelo 2	Modelo 1	Modelo 05
Train	0.5580887	0.6915998	0.7390952	0.7781568
Test	0.5471700	0.6711375	0.7117095	0.7467122
Nº hojas	**10**	**27**	**53**	**106**

Modelo 2

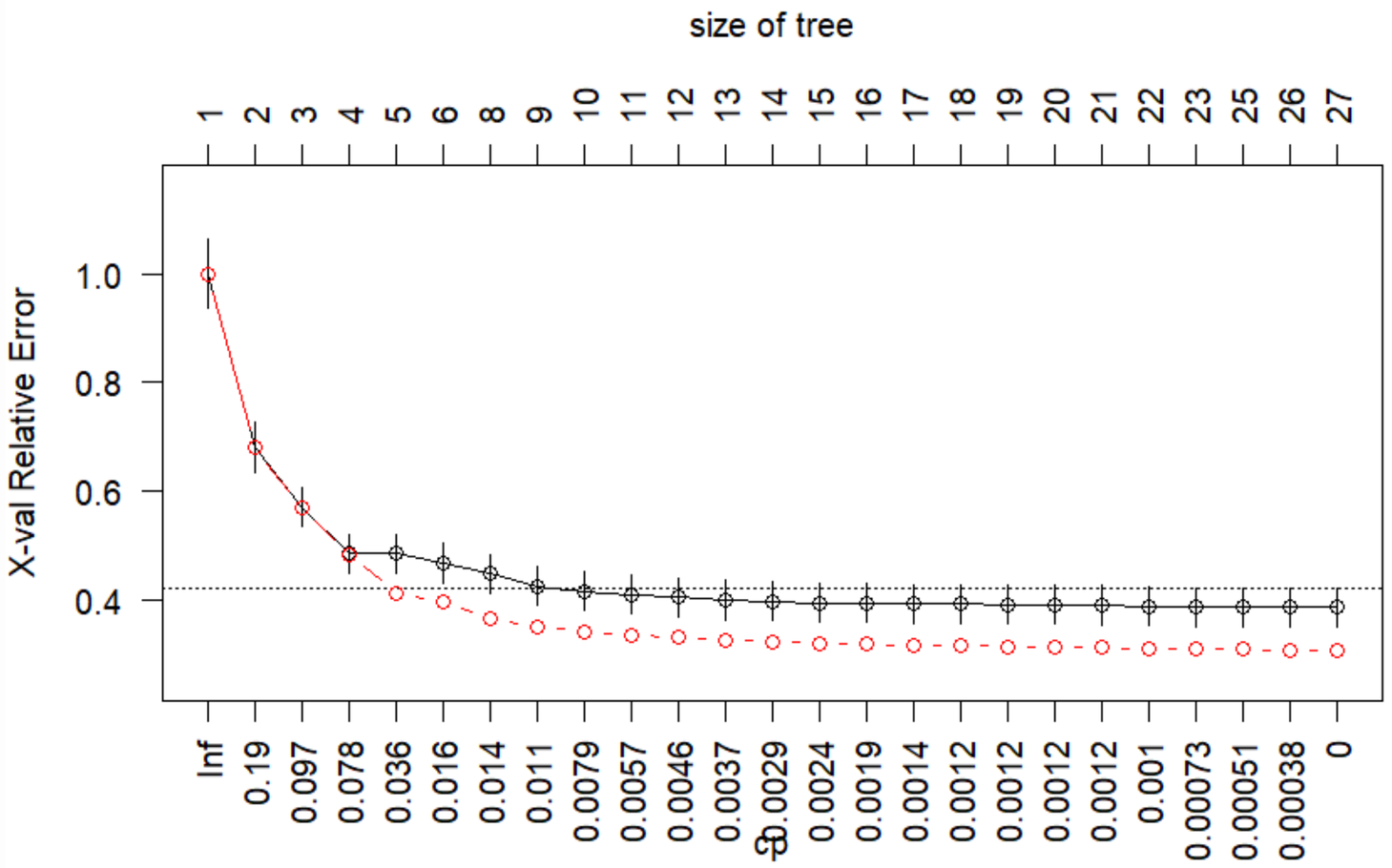
- Menos sobreajuste
- Buena predicción
  (para el número de hojas que posee)
- El más estable

PODA DEL ÁRBOL 2

En negro: Validación cruzada

En rojo: Entrenamiento

Tamaño óptimo del árbol: 10 hojas



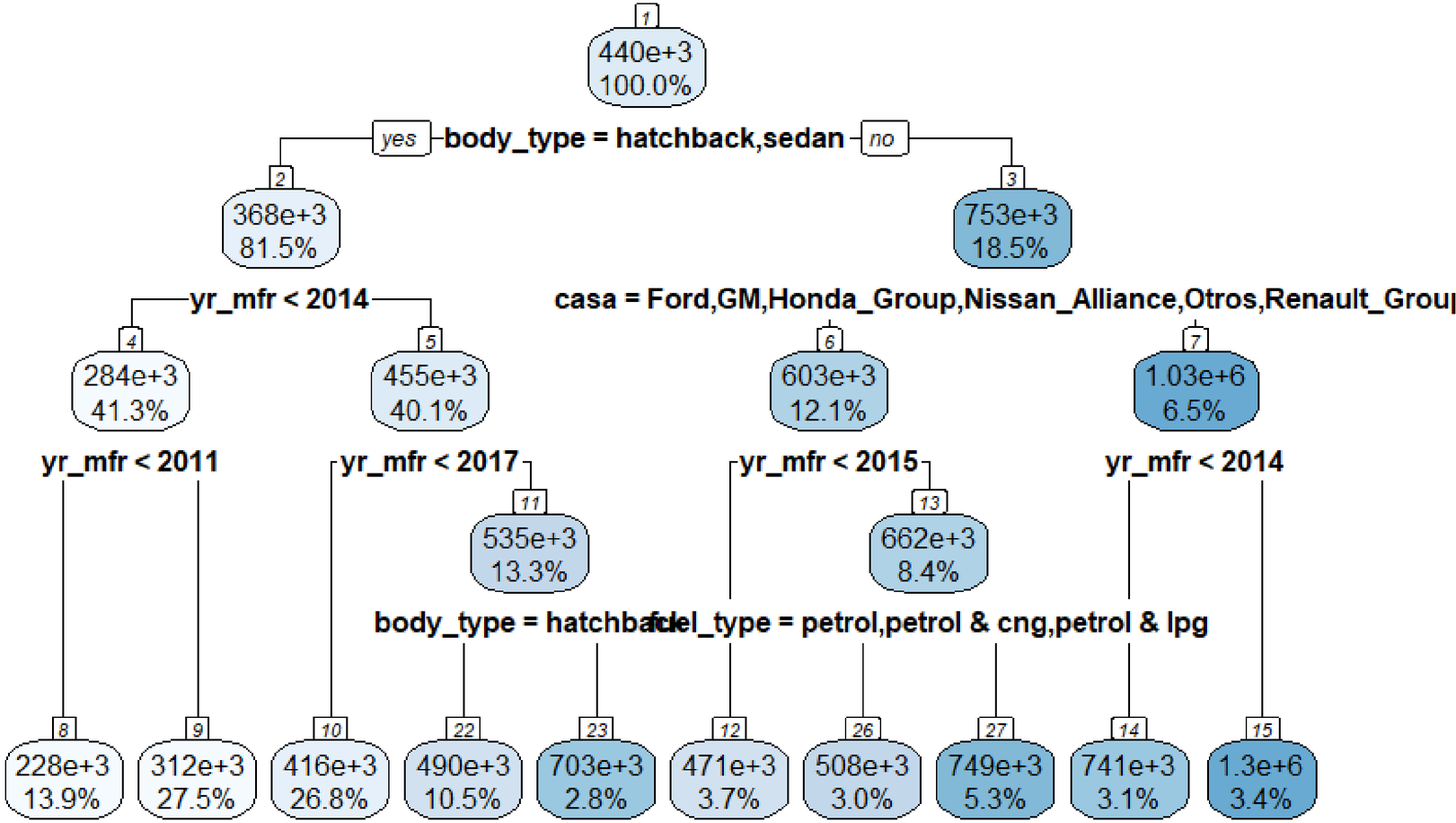
PODA DEL ÁRBOL 2

Tamaño óptimo del árbol: 10 hojas
Profundidad: 4

Nodo 8: coches con carrocería hatchback o sedan cuyo año de fabricación es inferior a 2011

Nodo 15: coches con carrocería diferente a hatchback o sedan cuyo año de fabricación es inferior a 2014

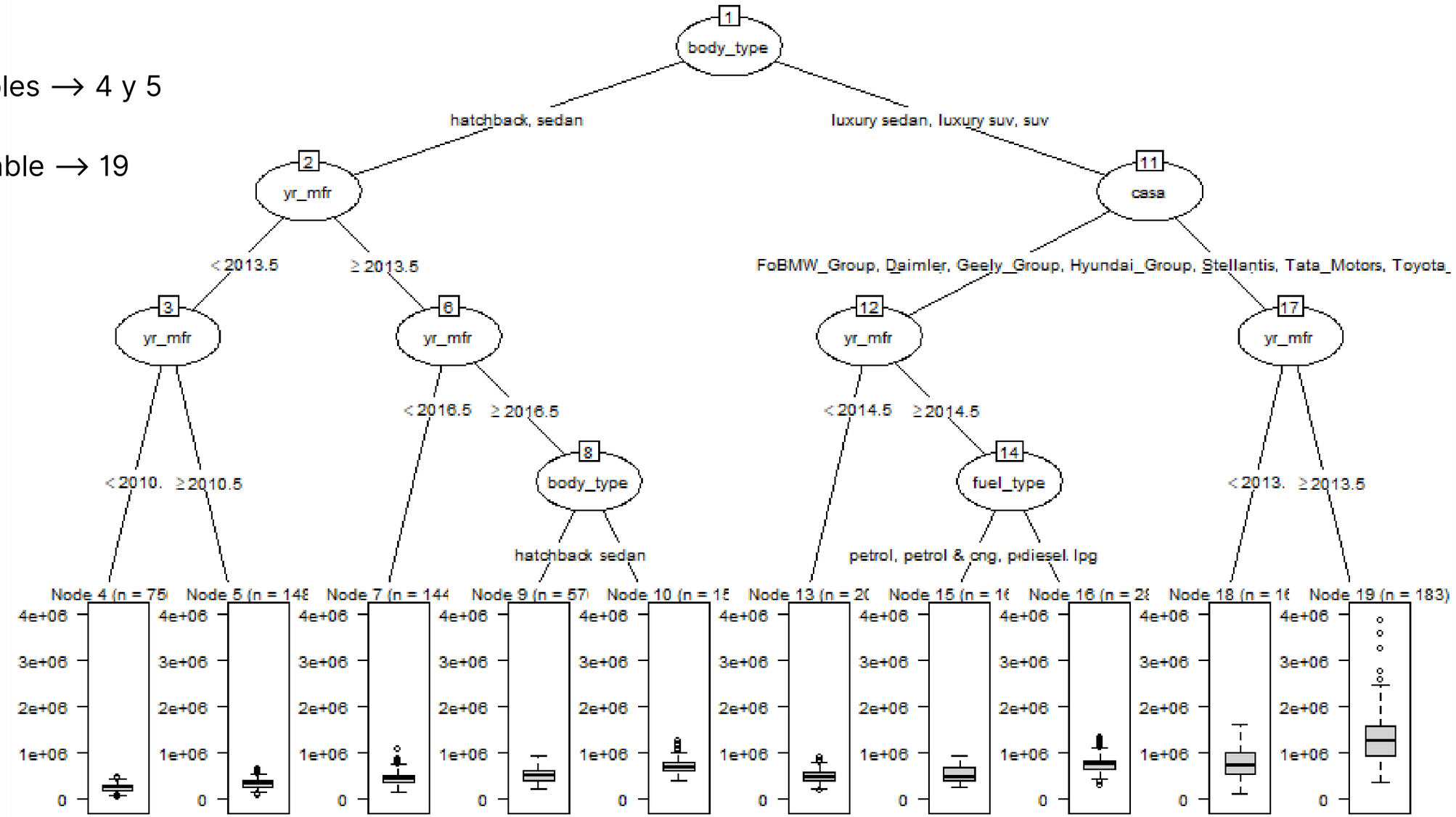
R2 en train y test de 0.6571219 y 0.6282152, respectivamente.



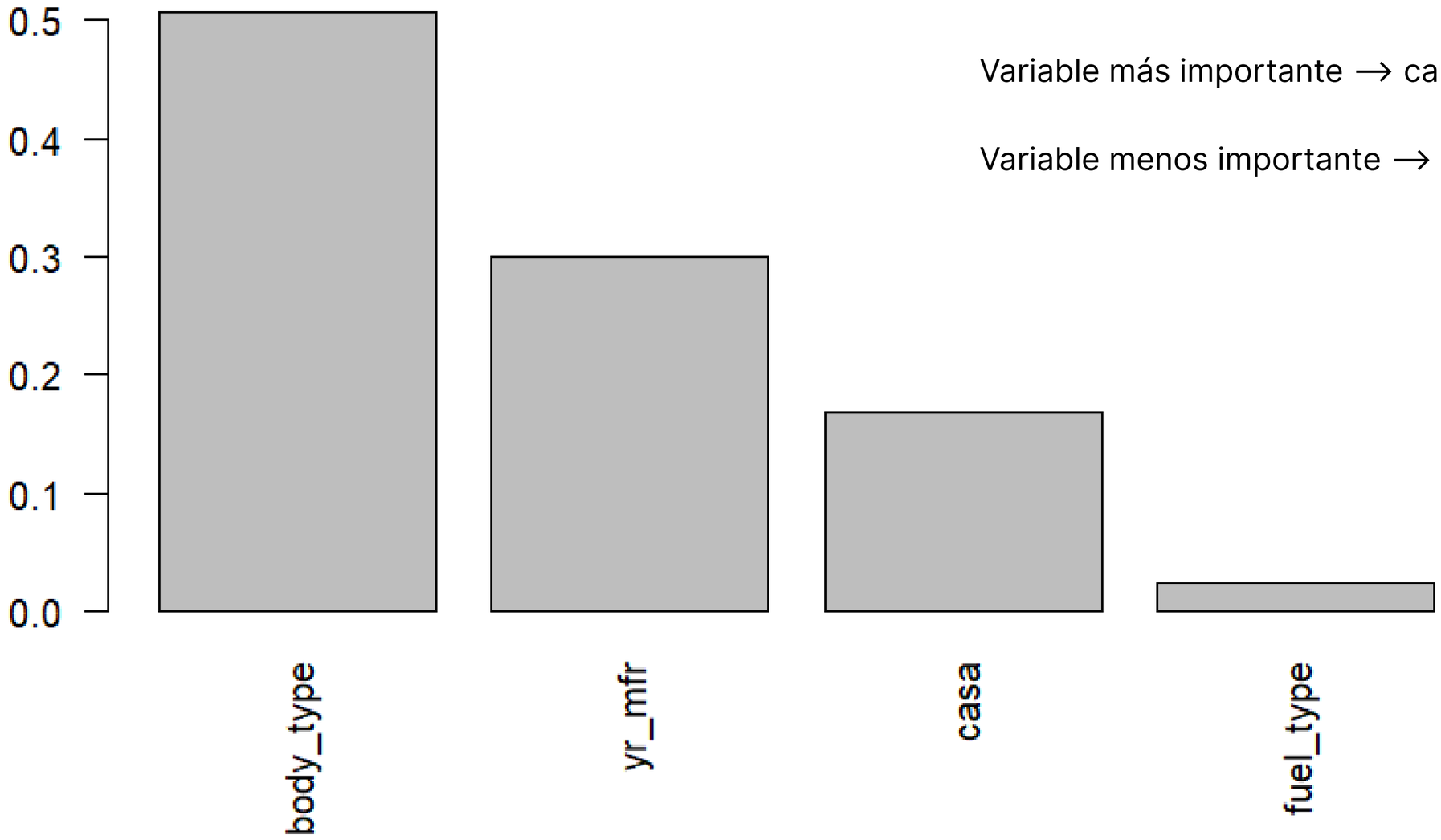
PODA DEL ÁRBOL 2

Nodos más fiables ⟶ 4 y 5

Nodo menos fiable ⟶ 19



IMPORTANCIA DE VARIABLES



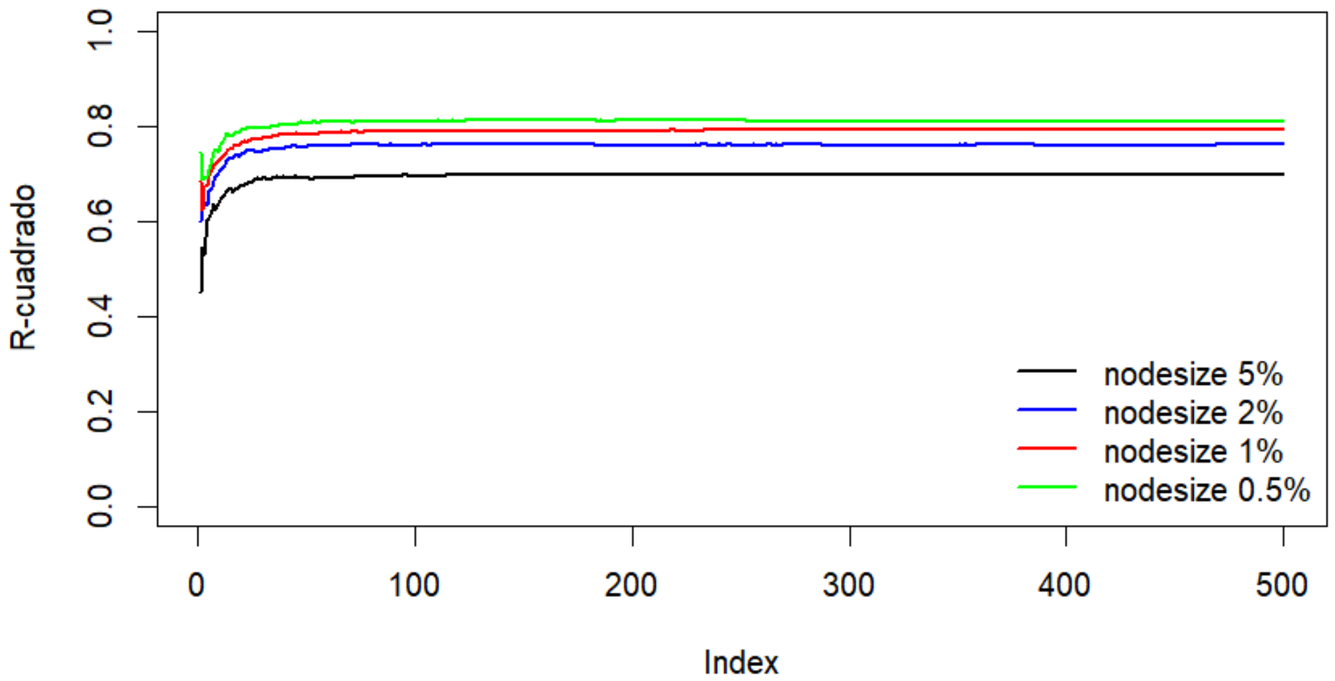
Variable más importante → carrocería

Variable menos importante → combustible

RANDOM FOREST

- Creación y selección de árboles

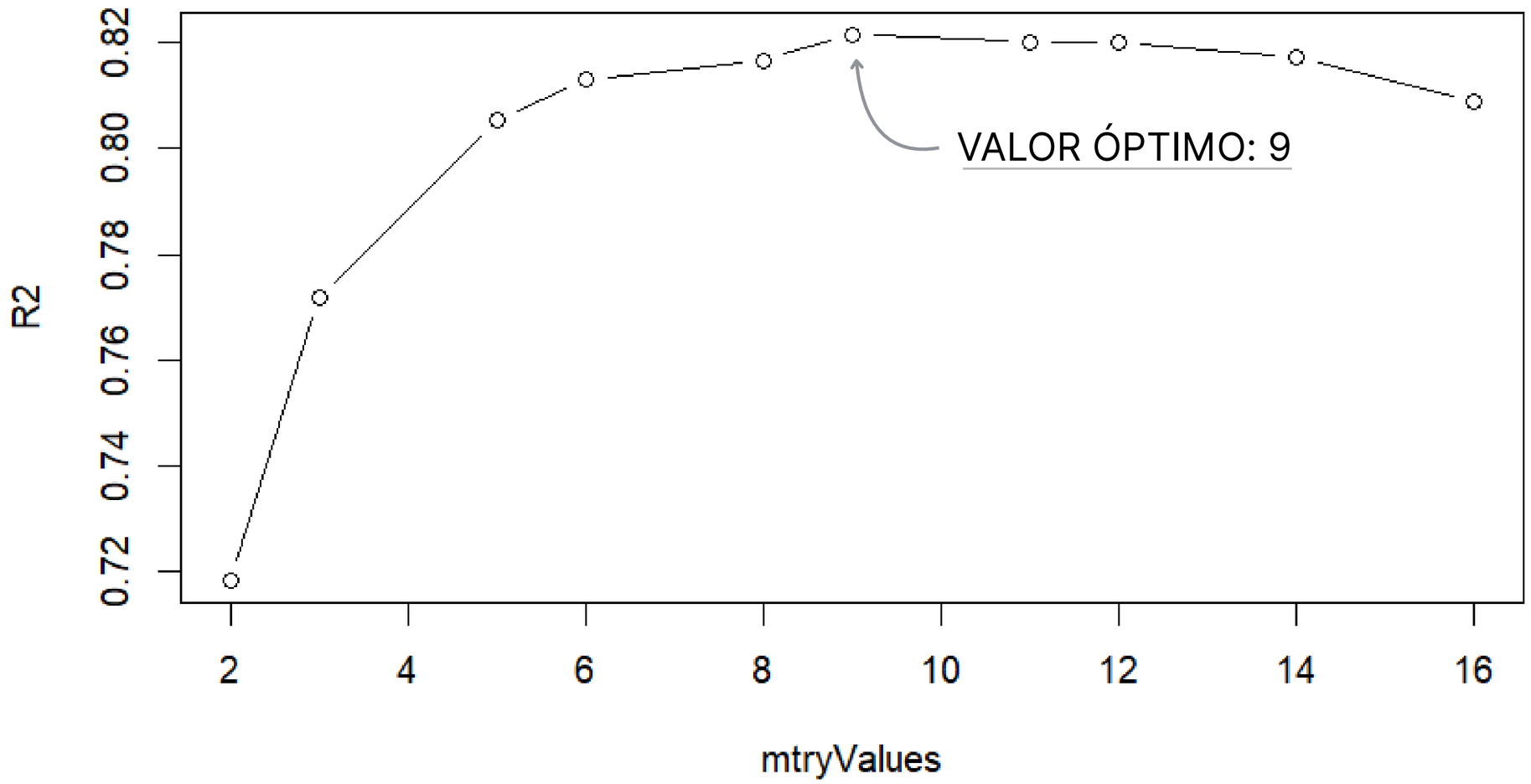
Mejora progresiva del R2



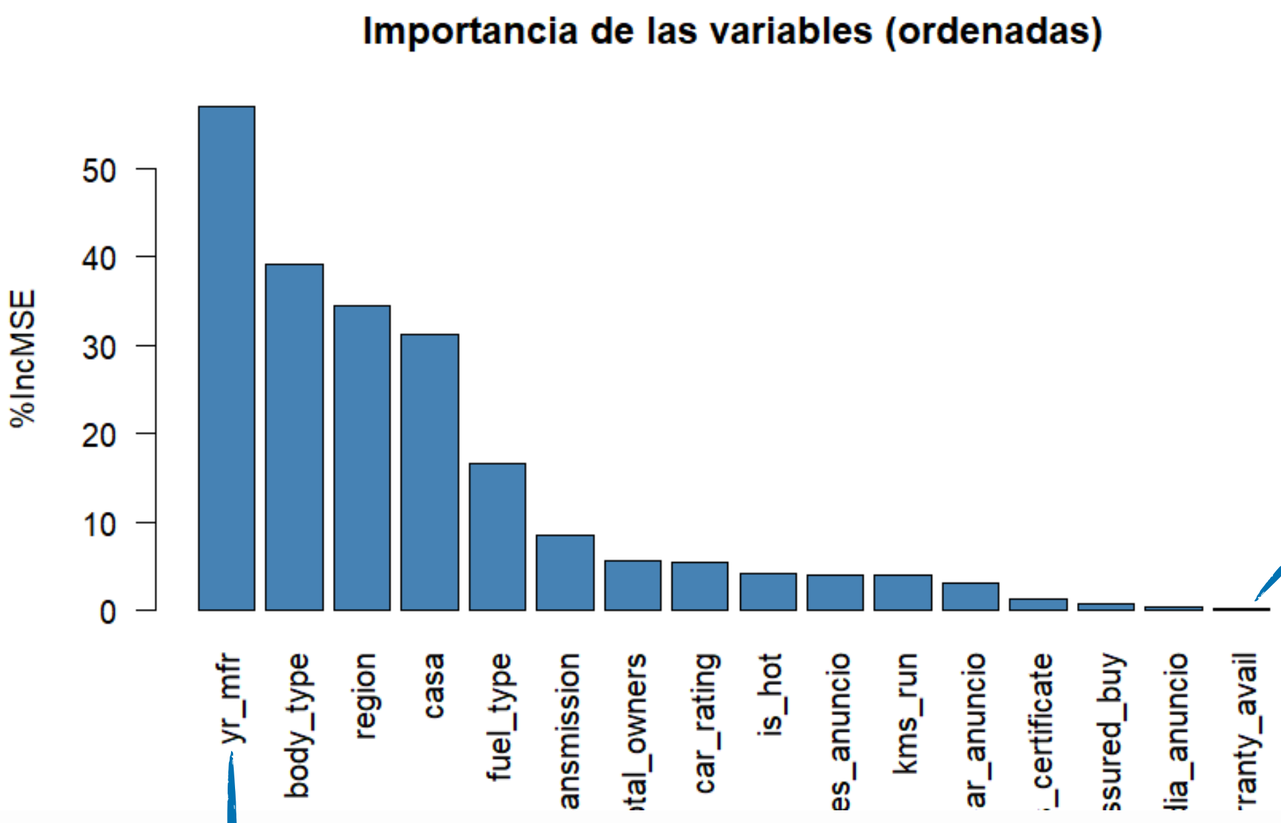
nodesize <chr>	R2_train <dbl>	R2_test <dbl>
5%	0.7517925	0.7341711
2%	0.8215611	0.7877590
1%	0.8647127	0.8168098
0.5%	0.8987878	0.8358511

Capacidad explicativa

MTRY


R2
0.72
0.74
0.76
0.78
0.80
0.82
2
4
6
8
10
12
14
16
mtryValues
VALOR ÓPTIMO: 9

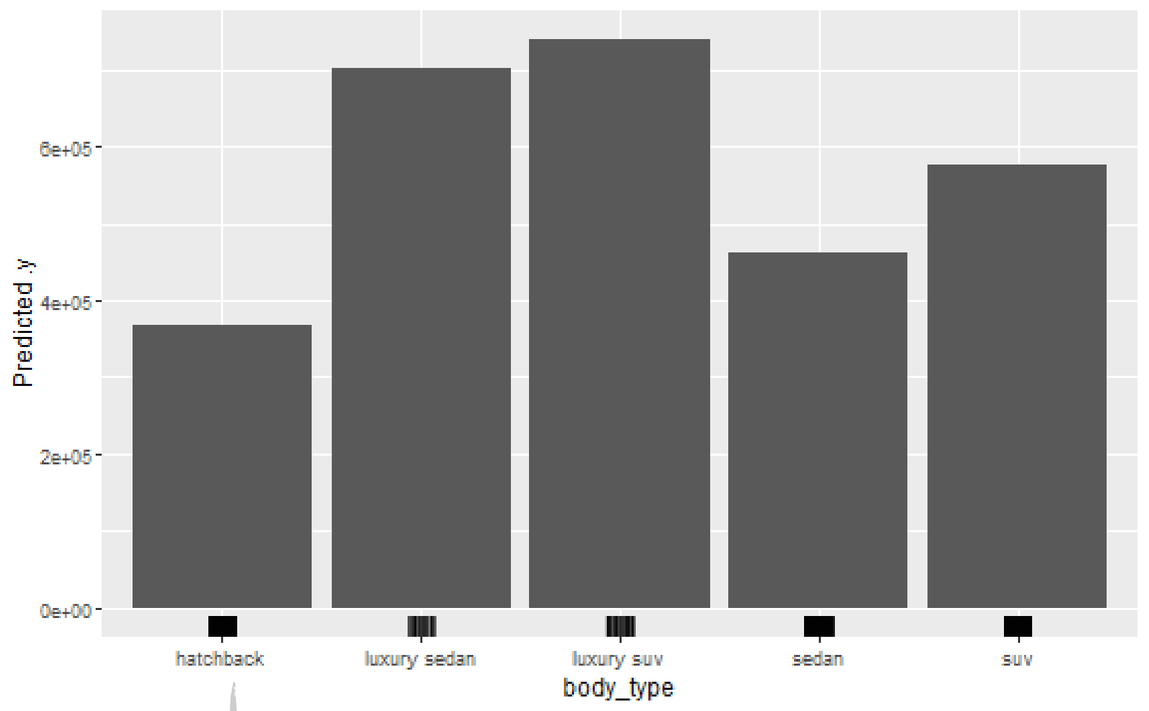
IMPORTANCIA DE VARIABLES



Disponibilidad de garantía

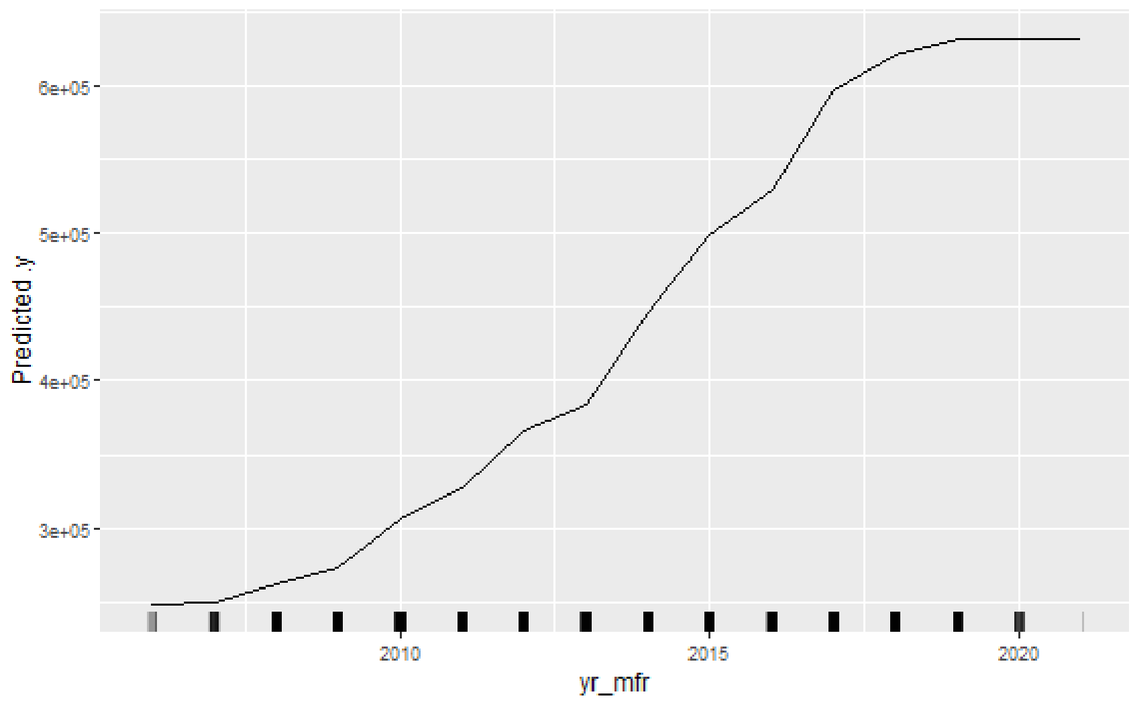
Año de fabricación

GRÁFICOS PDP



Variable *body_type*

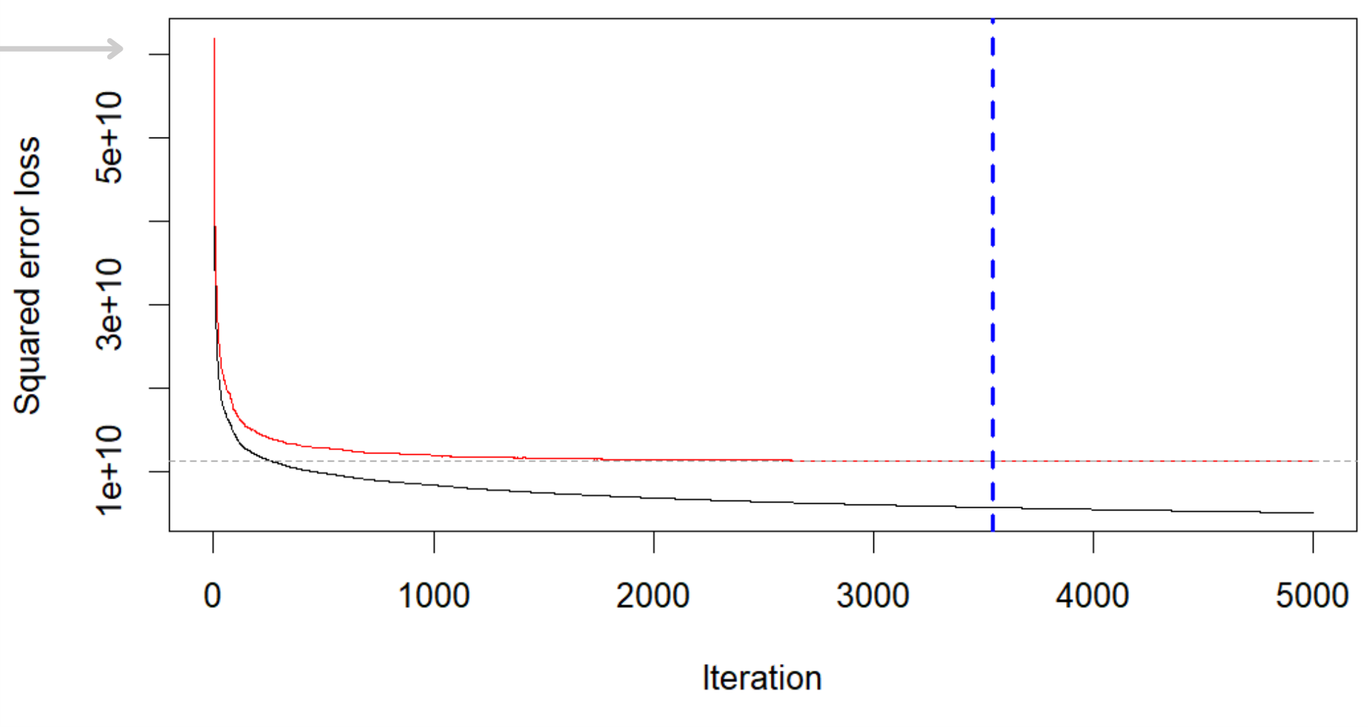
Variable *yr_mfr*



MODELO GRADIENT BOOSTING

- 5000 árboles ⟶ 3547 valor óptimo
- Tasa de aprendizaje (α) = 0.2
- Tamaño mínimo de la hoja del 5%
- Profundidad de 3

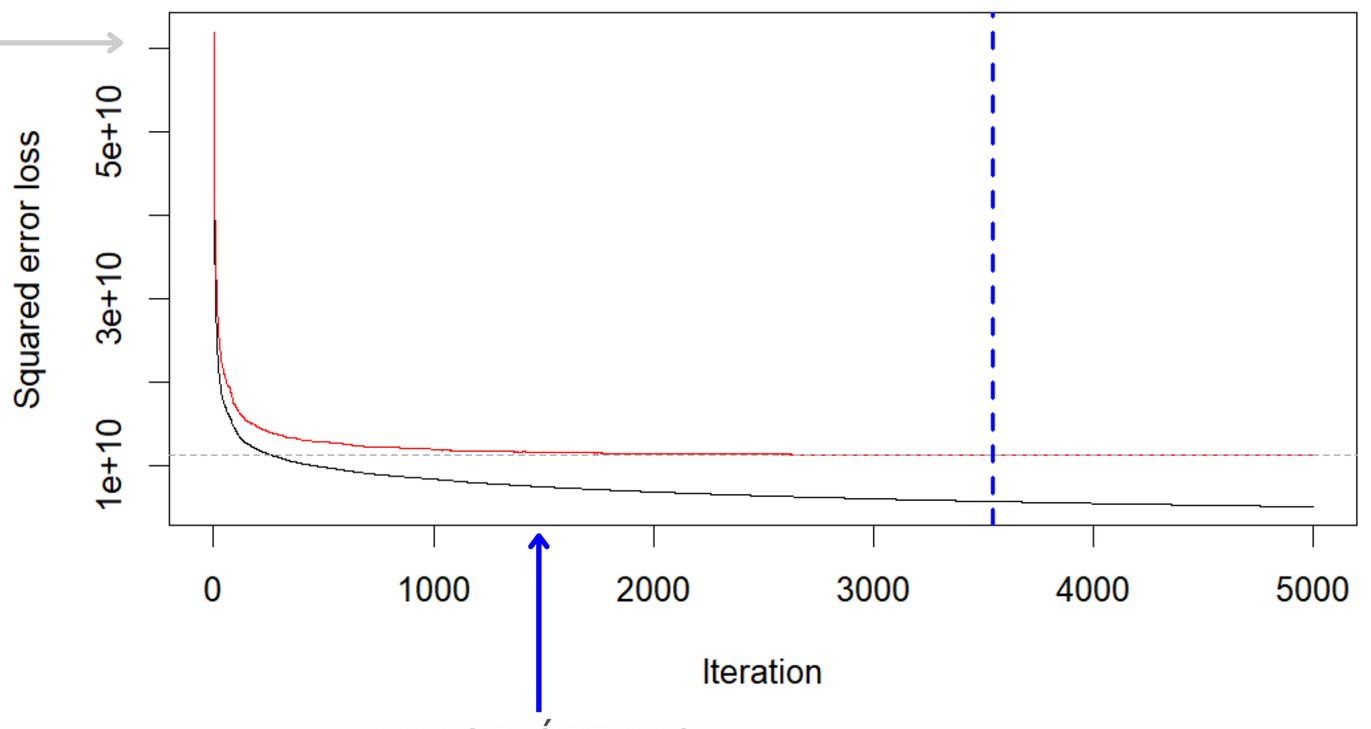
EVOLUCIÓN DEL MSE



MODELO GRADIENT BOOSTING

- 5000 árboles ⟶ 3547 valor óptimo
- Tasa de aprendizaje (α) = 0.2
- Tamaño mínimo de la hoja del 5%
- Profundidad de 3

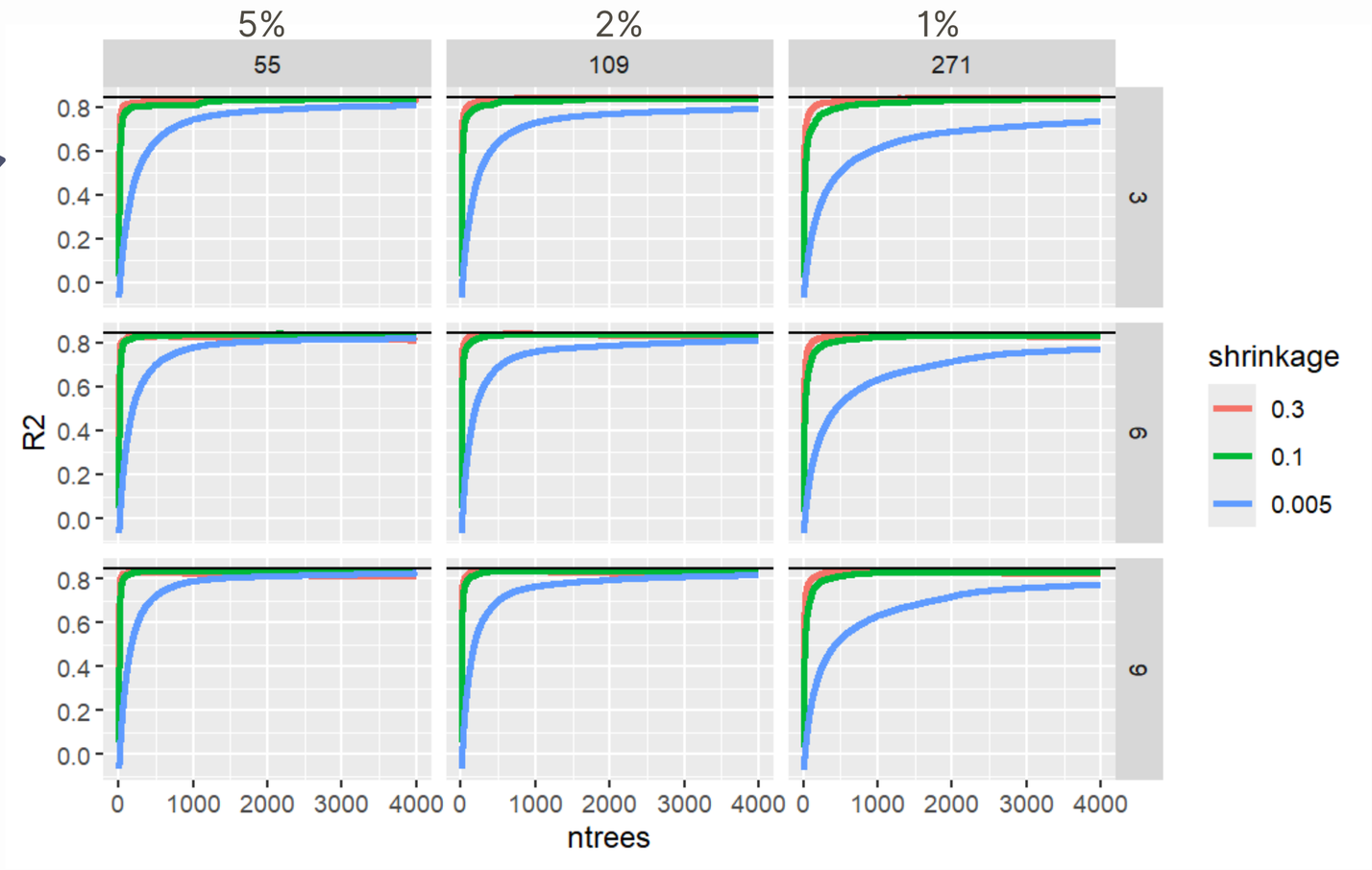
EVOLUCIÓN DEL MSE



VALOR ÓPTIMO

REJILLA

```
hyper_grid <- expand.grid(
  shrinkage = c(0.3, 0.1, 0.005),
  interaction.depth = c(3, 6, 9),
  n.minobsinnode = c(ceiling(0.01*nrow(datos_train)), ceiling(0.02*nrow(datos_train
)), ceiling(0.05*nrow(datos_train))))
```



- **tasas de aprendizaje:**
  0.3 0.1 0.005
- **profundidad:**
  3 6 9
- **tamaño min. de la hoja**
  1% 2% 5%

shrinkage <fctr>	interaction.depth <fctr>	n.minobsinnode <fctr>	ntrees <int>	R2 <dbl>
0.3	3	109	1693	0.8448879
0.3	3	271	1951	0.8438410
0.3	6	109	712	0.8432191
0.1	6	55	2144	0.8426307
0.1	6	109	3654	0.8414663
0.1	3	109	3941	0.8405196
0.3	3	55	1041	0.8396515
0.3	9	109	408	0.8395718
0.1	9	109	703	0.8385481
0.1	3	271	3985	0.8381349
0.1	3	55	3122	0.8376849
0.3	6	271	1014	0.8365547
0.1	9	55	1668	0.8359306
0.1	6	271	3349	0.8352036
0.3	9	55	167	0.8343287
0.3	9	271	810	0.8334448

- tasa de aprendizaje: 0.3
- profundidad: 9

- tamaño mínimo de la hoja: 5% (55 obs)
- número de árboles: 167

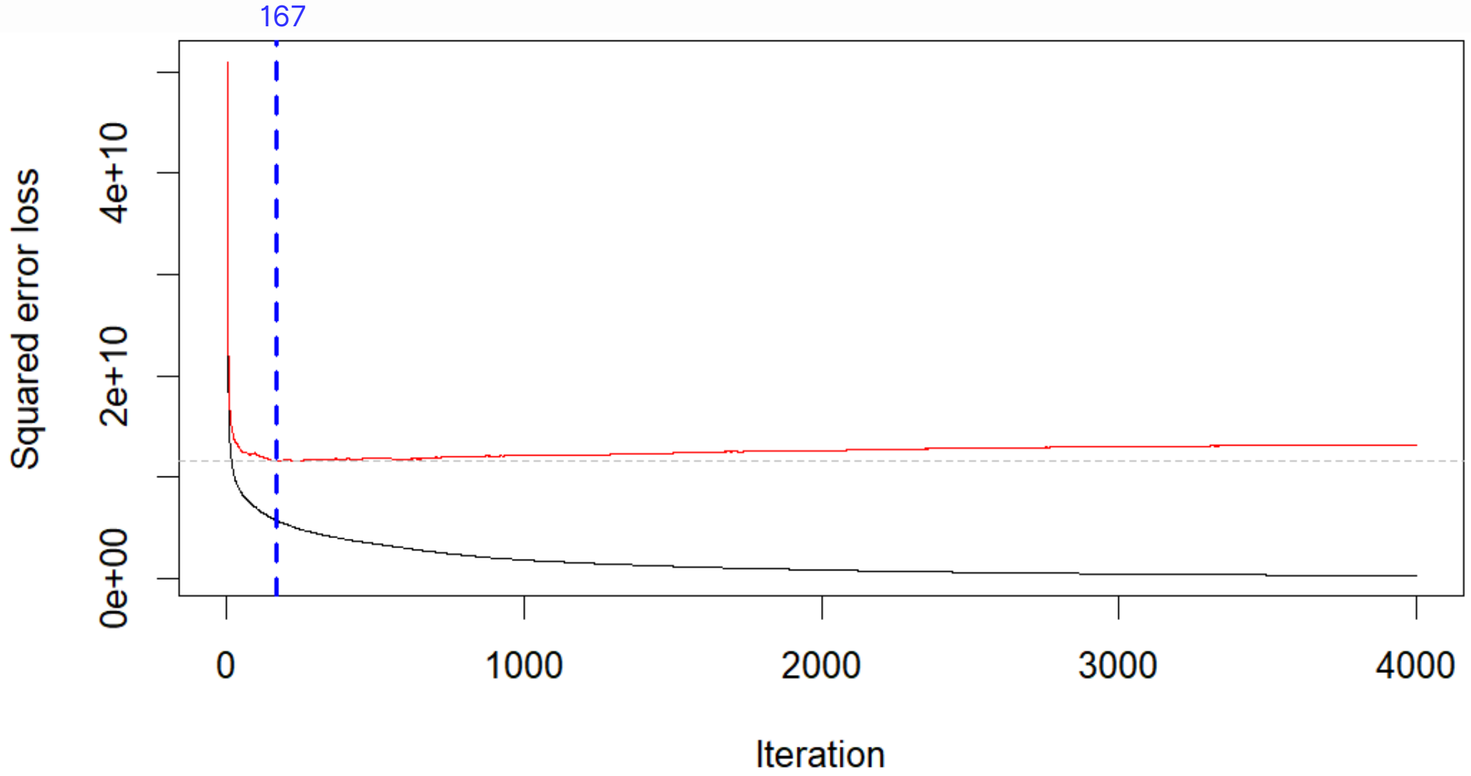
- $R^2 \approx 0.8343$

tasa de aprendizaje

- optComb<-which((hyper_grid$shrinkage==**0.3**)&(hyper_grid$interaction.depth==**9**)&(hyper_grid$n.minobsinnode==**55**))

profundidad

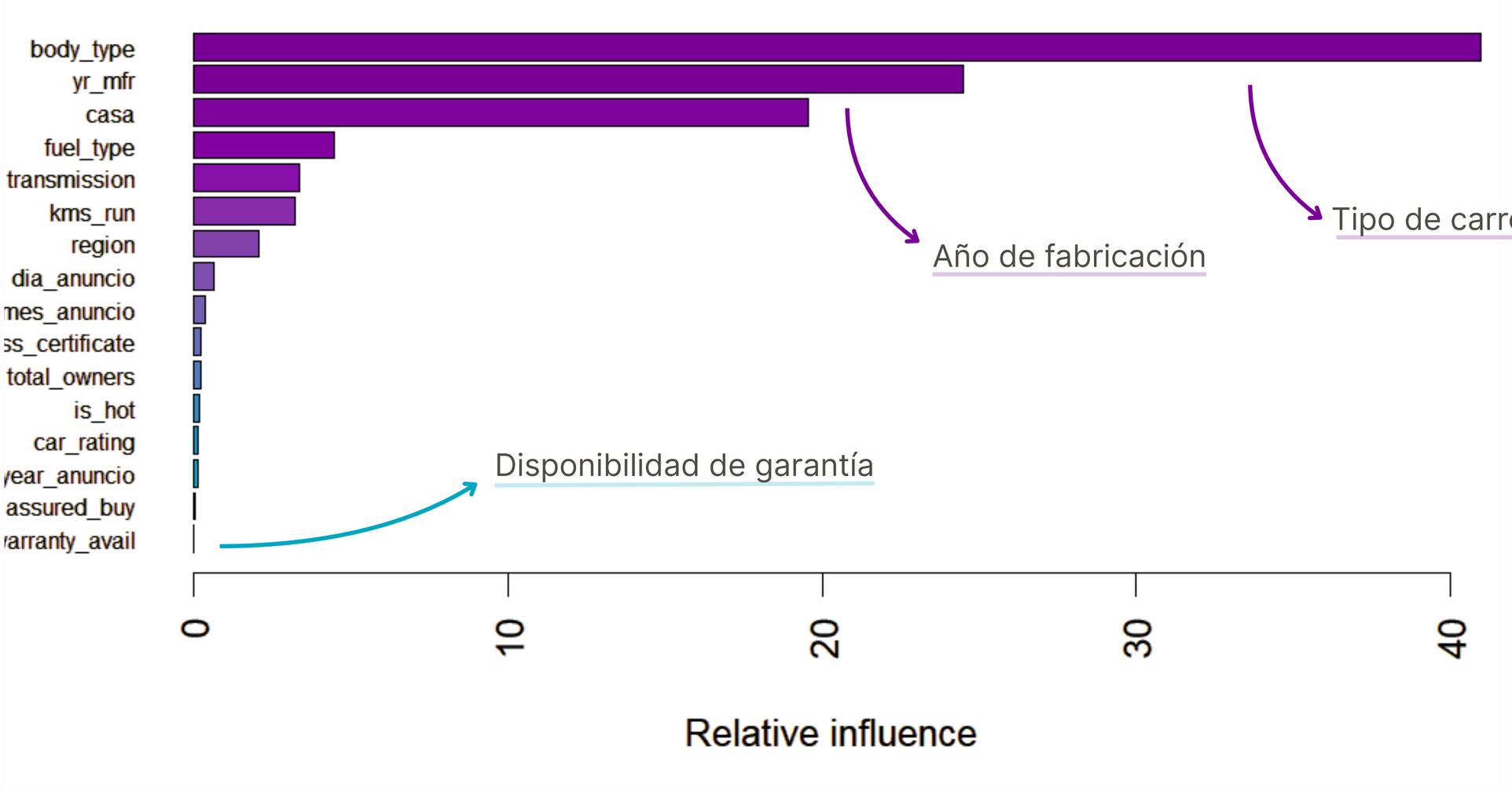
tamaño mínimo de la hoja



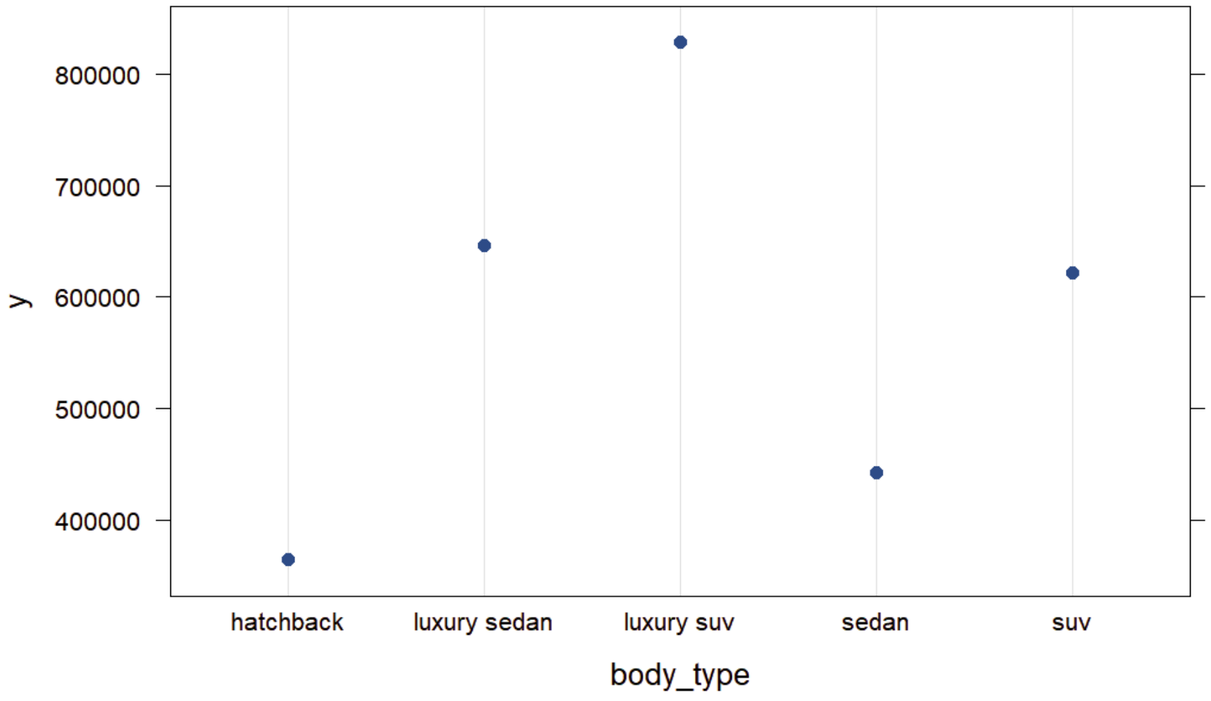
EVALUACIÓN DEL MODELO

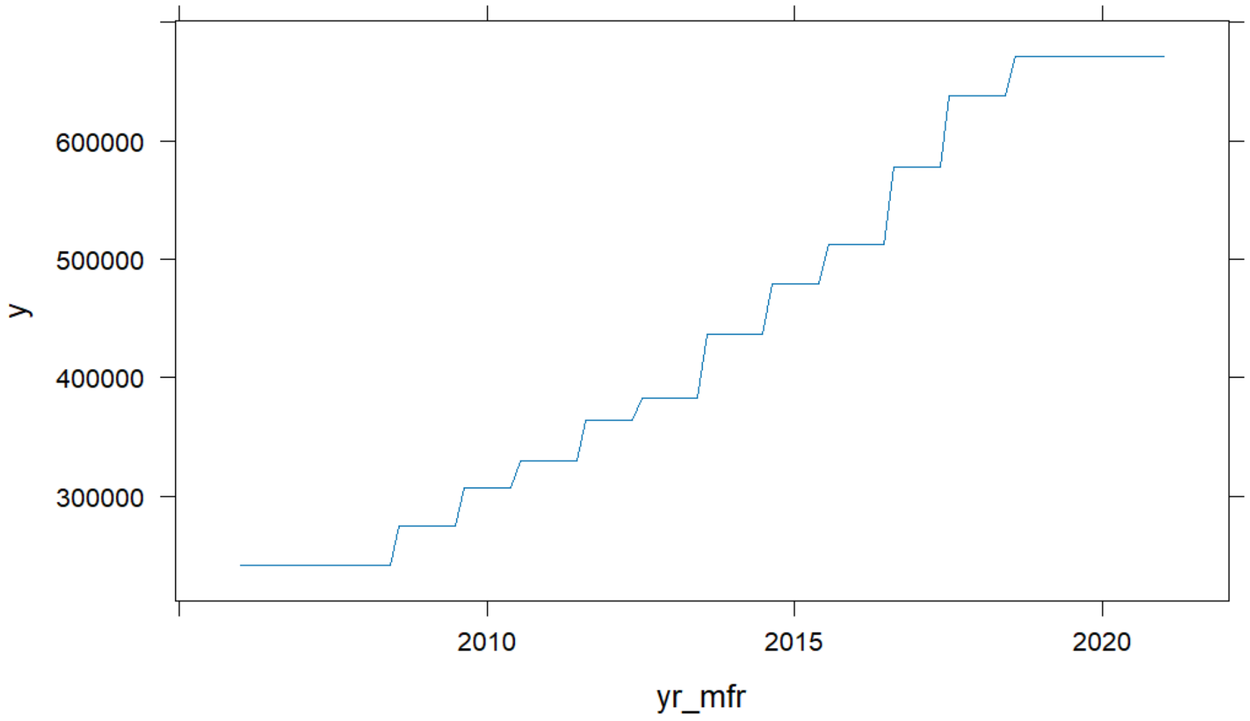
- Entrenamiento: 0.9150686
- Prueba: 0.8621249

IMPORTANCIA DE VARIABLES



GRÁFICOS PDP





CONCLUSIÓN

MODELO SELECCIONADO:

Gradient Boosting